

1 Appendix Model estimation with deep neural nets

Call $x(kf, kc, kd, ks, T, \tau)$ the vector of state variables: final capital, clean capital, dirty capital, stranded capital, temperature and synthetic time. Temperature is proportional to cumulative emissions S , $T = \zeta S$. We will write first order conditions of S rather than T , but T has a better range as an input in a neural net (S is 3 magnitudes larger). We will train a deep neural net which takes in the state variables and gives as output the policies: optimal decision variables (investments), shadow prices and value function $\mathcal{N}(x) = (if, ic, id, \lambda^{ks}, \lambda^S, v)$.

We express all variables in units of effective labour, for example consumption per capita is $c = \frac{C}{A_0 L_0 e^{(g+g_L)t}}$ and dirty capital $kd = \frac{Kd}{A_0 L_0 e^{(g+g_L)t}}$.

Production is,

$$y = (\varphi_f k f^{\epsilon_f} + \varphi_E E^{\epsilon_f})^{\alpha/\epsilon_f} \underbrace{e^{-\frac{\gamma}{2}\zeta^2 S^2}}_{\Omega} \quad (1)$$

With energy $E = (\varphi_c k c^\epsilon + \varphi_d (kd + \xi)^\epsilon)^{1/\epsilon}$ and ξ a very small number to avoid infinite marginal abatement costs at zero emissions. It also ensures a stable temperature in the steady state.

Stranding corresponds to negative dirty investment. Consumption equals production y minus investment (provided they are positive) minus the stranding adjustment costs.

$$c = y - if - ic - sp_{\varepsilon_1}(id) - 1_{id < 0} \chi_s \frac{id^2}{kd}, \quad (2)$$

where sp_{ε_1} is the soft-plus function $\ln(1 + \exp(\varepsilon_1 id)) / \varepsilon_1$.

Dirty capital, stranding and emissions (separate appendix)

Dirty capital is heterogeneous in its carbon intensity, which is indexed by i . Define the function $\psi(Kd_i)$ as the carbon intensity of dirty capital of type i . The carbon intensity is ranked from the highest to the lowest, that is, decreasing in i . ¹We fit this function as an exponential function $\psi(Kd_i) = e^{-g_\psi t} (\psi_1 e^{-\psi_2 Kd_i} + \psi_3)$, with g_ψ representing the autonomous energy efficiency improvement, which converges to $g_L + g$, that is, $g_\psi = g_L + g - (g_L + g - g_{\psi 0}) (e^{-gg_\psi t})$. Integrating this function gives the emissions. Integrating this function over dirty capital between 0 and Kd_0 gives initial emissions

$$P_0 = \frac{\psi_1}{\psi_2} (1 - e^{-\psi_2 Kd_0}) + \psi_3 Kd_0. \quad (3)$$

¹The simplest approach is to use yearly emissions divided by the value of tangible assets E/V . In this case we assume that the capital decays at constant rate σ independent of the age of the assets and that the remaining emissions of any asset is E/σ . In reality, the remaining emissions of an asset are highly dependent on the age and are better approximated by $E(LT - age)$. So we will use $\frac{E}{V} \frac{(LT - age)}{1/\sigma}$ as the forward-looking carbon intensity over the remaining lifetime of the asset. V in this formula is the value of the asset at a given age, for which we can use the accounting value or else an age-dependent discounted cash flow formula. This forward-looking carbon intensity corresponds to $\frac{\sigma}{SCCE}$, where SCCE is the Stranding Cost of Cumulative Emissions.

Dirty capital decays at rate δ .² Emissions in subsequent periods are

$$P_t = A_0 L_0 \exp((g + g_L - g_\psi)t) \left\{ \frac{\psi_1}{\psi_2} \frac{kd_t + ks_t}{Kd_0} \left[\exp\left(-\psi_2 Kd_0 \frac{ks_t}{kd_t + ks_t}\right) - \exp(-\psi_2 Kd_0) \right] + \psi_3 kd_t \right\} \quad (4)$$

The marginal effect of dirty capital on emissions is

$$\begin{aligned} \frac{\partial P}{\partial kd} = & A_0 L_0 \exp((g + g_L - g_\psi)t) \\ & \left\{ \frac{\psi_1}{\psi_2} \exp\left(-\psi_2 Kd_0 \frac{ks_t}{kd_t + ks_t}\right) \left[\frac{1}{Kd_0} + \psi_2 \frac{ks_t}{kd_t + ks_t} \right] - \frac{\psi_1}{\psi_2 Kd_0} \exp(-\psi_2 Kd_0) + \psi_3 \right\} \end{aligned} \quad (5)$$

For stranding, the marginal carbon intensity is much larger, the decrease in emission from stranding one unit of capital is

$$\frac{\partial P}{\partial kd} - \frac{\partial P}{\partial ks} = A_0 L_0 \exp(g + g_L - g_\psi t) \left(\psi_1 \exp\left(-\psi_2 Kd_0 \frac{ks_t}{kd_t + ks_t}\right) + \psi_3 \right), \quad (6)$$

which uses the following result

$$\begin{aligned} \frac{\partial P}{\partial ks} = & -A_0 L_0 \exp((g + g_L - g_\psi)t) \left\{ \underbrace{-\frac{\psi_1}{\psi_2} \frac{1}{Kd_0} \left[\exp\left(-\psi_2 Kd_0 \frac{ks_t}{kd_t + ks_t}\right) - \exp(-\psi_2 Kd_0) \right]}_{\cong \psi} \right. \\ & \left. + \psi_1 \frac{kd_t}{kd_t + ks_t} \exp\left(-\psi_2 Kd_0 \frac{ks_t}{kd_t + ks_t}\right) \right\} \end{aligned}$$

$\frac{\partial P}{\partial ks}$ shrinks rapidly over time, even after centuries despite the exponential positive rate $g + g_L - g_\psi$, because the effect of stranding is smaller when ks is large. Also, $\frac{ks}{kd + ks}$ converges to 1. This is a side-effect of the soft-plus function but it is irrelevant because $\frac{\partial P}{\partial ks}$ only affects λ^{ks} which is irrelevant after stranding in the first years.

The elasticity of substitution between clean and dirty energy increases over time as a result of endogenous and exogenous technical change, combined as follows

$$\sigma = \frac{\sigma_0}{(1 - \omega)M^{-\varrho} + \omega e^{-g_\epsilon t}} \Leftrightarrow \epsilon = 1 - \frac{(1 - \omega)M^{-\varrho} + \omega e^{-g_\epsilon t}}{\sigma_0} \quad (7)$$

with σ_0 the initial elasticity of substitution. Endogenous technical change is driven by cumulative abatement M ,

$$M = \frac{M_0 + P_{BAU}t - (S - S_0)}{M_0} \quad (8)$$

where P_{BAU} are constant initial emissions and S are cumulative emissions.

²Due to this decay, the carbon intensity of a given type of dirty capital i at a future time t equals the intensity of the initial function with the decay-scaled argument $\psi\left(Kd_i \frac{Kd_0}{Kd_t + Ks_t}\right)$. Over periods with negative investment, $\frac{Kd_0}{Kd_t + Ks_t} = e^{-\delta \Delta t}$, over periods of positive investment, the formula implies that the newly invested capital is of the average carbon intensive type. As a result, the least carbon intensive type of dirty capital is always $\varphi_1 e^{-\psi_2 Kd_0}$

Synthetic time (t is years, not the number of periods)

$$\tau = 1 - e^{-g_\tau t} \Leftrightarrow t = \frac{-\ln(1 - \tau)}{g_\tau} \Leftrightarrow \frac{\partial t}{\partial \tau} = \frac{1}{g_\tau(1 - \tau)} \quad (9)$$

Equations of motion (all flows are expressed in yearly units)

$$\begin{aligned} 0 &= -k f_{t+\Delta t} + (1 - \delta - g_L - g)^{\Delta t} k f_t + i f \Delta t - \chi \frac{i f^2}{k f} \Delta t + \sigma_{kf} k f_t dW^f \sqrt{\Delta t} \\ 0 &= -k c_{t+\Delta t} + (1 - \delta - g_L - g)^{\Delta t} k c_t + i c \Delta t - \chi \frac{i c^2}{k c} \Delta t + \sigma_{kc} k c_t dW^c \sqrt{\Delta t} \\ 0 &= -k d_{t+\Delta t} + (1 - \delta - g_L - g)^{\Delta t} k d_t + i d \Delta t - 1_{id>0} \chi \frac{i d^2}{k d} \Delta t + \sigma_{kd} k d_t dW^d \sqrt{\Delta t} \\ 0 &= -k s_{t+\Delta t} + (1 - \delta - g_L - g)^{\Delta t} k s_t + s p_{\varepsilon_1, \varepsilon_2}(-i d) \Delta t \\ 0 &= -S_{t+\Delta t} + S_t + P_t(kd, ks, \tau) \Delta t + \sigma_S S_t dW^S \sqrt{\Delta t} \end{aligned}$$

The recursive problem reads as follows³

$$v(x_t) = \max_{\{if_t, ic_t, id_t\}} \left\{ \frac{c^{1-\eta}}{1-\eta} \Delta t + \frac{e^{-(r-g)\Delta t}}{1-\eta} \left\{ \mathbb{E} \left[((1-\eta) v(x_{t+\Delta t}))^{\frac{1-RR A}{1-\eta}} \right] \right\}^{\frac{1-\eta}{1-RR A}} \right\}, \quad (10)$$

Instead of acknowledging that $x_{t+\Delta t} = g(x_t, if, ic, id)$ via the equations of motion, it is more convenient to consider $x_{t+\Delta t}$ as 6 extra choice variables while adding the equations of motions as constraints. The constrained maximization then corresponds to the following Lagrangian:

$$\begin{aligned} \mathcal{L} = & \max_{if_t, ic_t, id_t, x_{t+\Delta t}} \left\{ \frac{c_t^{1-\eta}}{1-\eta} \Delta t - e^{-(r-g)\Delta t} \left[\mathbb{E} \left[(-v(x_{t+\Delta t}))^{\frac{1-RR A}{1-\eta}} \right] \right]^{\frac{1-\eta}{1-RR A}} \right. \\ & \left. + \sum_t \sum_{i=\{kf, kc, kd, Ks, S\}} \lambda_t^i [Constraint^i] \right\} \end{aligned} \quad (11)$$

The first order conditions for the maximization of the Lagrangian are⁴

³Since the value function is negative for $\eta < 1$, we switch the sign of v by adding the factor $1 - \eta$ in the expectation and $\frac{1}{1-\eta}$ outside the expectation (note that both powers add up to 1). In the code, since eta is smaller than one we replace $1 - \eta$ by -1 .

⁴The derivative of the softplus function $sp_\varepsilon(id)$ is the sigmoid function $\varsigma(id\varepsilon) = \frac{1}{1+\exp(-id\varepsilon)}$

$$\begin{aligned}
\frac{\partial \mathcal{L}_t}{\partial i f_t} &= 0 = c^{-\eta} (-1) + \lambda^{kf} \left(1 - 2\chi \frac{if}{kf} \right), \\
\frac{\partial \mathcal{L}_t}{\partial i c_t} &= 0 = c^{-\eta} (-1) + \lambda^{kc} \left(1 - 2\chi \frac{ic}{kc} \right), \\
\frac{\partial \mathcal{L}_t}{\partial i d_t} &= 0 = c^{-\eta} \left(-\varsigma(\varepsilon_1 id) - 1_{id < 0} 2\chi_s \frac{id}{kd} \right) \\
&\quad + \lambda^{kd} \left(1 - 1_{id > 0} 2\chi \frac{id}{kd} \right) + \lambda^{ks} \varsigma((-id + \varepsilon_2)\varepsilon_1) (-1), \\
\frac{\partial \mathcal{L}_t}{\partial k f_{t+\Delta t}} &= 0 = e^{-(r-g)\Delta t} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{1-RR A}{1-\eta}} \right]^{\frac{RR A - \eta}{1-RR A}} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{\eta-RR A}{1-\eta}} v_{kf}(x_{t+1}) \right] - \lambda_t^{kf}, \\
\frac{\partial \mathcal{L}_t}{\partial k d_{t+\Delta t}} &= 0 = e^{-(r-g)\Delta t} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{1-RR A}{1-\eta}} \right]^{\frac{RR A - \eta}{1-RR A}} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{\eta-RR A}{1-\eta}} v_{kd}(x_{t+1}) \right] - \lambda_t^{kd}, \\
\frac{\partial \mathcal{L}_t}{\partial k c_{t+\Delta t}} &= 0 = e^{-(r-g)\Delta t} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{1-RR A}{1-\eta}} \right]^{\frac{RR A - \eta}{1-RR A}} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{\eta-RR A}{1-\eta}} v_{kc}(x_{t+1}) \right] - \lambda_t^{kc}, \\
\frac{\partial \mathcal{L}_t}{\partial k s_{t+\Delta t}} &= 0 = e^{-(r-g)\Delta t} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{1-RR A}{1-\eta}} \right]^{\frac{RR A - \eta}{1-RR A}} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{\eta-RR A}{1-\eta}} v_{ks}(x_{t+1}) \right] - \lambda_t^{ks}, \\
\frac{\partial \mathcal{L}_t}{\partial S_{t+\Delta t}} &= 0 = e^{-(r-g)\Delta t} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{1-RR A}{1-\eta}} \right]^{\frac{RR A - \eta}{1-RR A}} \mathbb{E} \left[(-v(x_{t+1}))^{\frac{\eta-RR A}{1-\eta}} v_S(x_{t+1}) \right] - \lambda_t^S,
\end{aligned}$$

The first three equations are used to substitute out $\lambda^{kf}, \lambda^{kc}, \lambda^{kd}$ from the solutions. The sum of the errors of the other first order conditions, together with the error of the Bellman equation, will constitute the loss function.

To find the partial derivatives of the value function in the loss function, we apply the envelope theorem to the Lagrangian, acknowledging that the Lagrangian represents the value function,⁵

⁵The Lagrangian is maximized over x_{t+1} so that the derivatives wrt to x_{t+1} are zero. This is not the case for x_t . because the Lagrangian is in fact the value function at time t, which depends on x_t , so the derivatives are not zero, but rather the derivatives of the value function.

$$\begin{aligned}
v_{kf} = \frac{\partial \mathcal{L}_t}{\partial k f_t} &= c^{-\eta} \alpha y \underbrace{\frac{\varphi_{kf} k f^{\epsilon_f - 1}}{\varphi_f k f^{\epsilon_f} + \varphi_E E^{\epsilon_f}}}_{y_{kf}} \Delta t \\
&\quad + \lambda^{kf} \left[(1 - \delta - g_L - g)^{\Delta t} + \chi \frac{i f^2}{k f^2} \Delta t \right] \\
v_{kc} = \frac{\partial \mathcal{L}_t}{\partial k c_t} &= c^{-\eta} \alpha y \underbrace{\frac{\overbrace{\varphi_E E^{\epsilon_f - 1}}^{y_E}}{\varphi_f k f^{\epsilon_f} + \varphi_E E^{\epsilon_f}}}_{y_{kc} = \frac{\alpha y \varphi_E \varphi_c k c^{\epsilon - 1} E b a s e^{\epsilon_f / \epsilon - 1}}{y_{base}}} \underbrace{\overbrace{E^{1 - \epsilon}}^{E_{kc}} \varphi_c k c^{\epsilon - 1}}_{E b a s e^{1 / \epsilon - 1}} \Delta t \\
&\quad + \lambda^{kc} \left[(1 - \delta - g_L - g)^{\Delta t} + \chi \frac{i c^2}{k c^2} \Delta t \right] \\
v_{kd} = \frac{\partial \mathcal{L}_t}{\partial k d_t} &= c^{-\eta} \left[\underbrace{\alpha y \frac{\varphi_E E^{\epsilon_f - 1}}{\varphi_f k f^{\epsilon_f} + \varphi_E E^{\epsilon_f}} \overbrace{E^{1 - \epsilon} \varphi_d (k d + \xi)^{\epsilon - 1}}^{E_{kd}}}_{y_{kd}} \Delta t + 1_{id < 0} \chi_s \frac{i d^2}{k d^2} \Delta t \right] \\
&\quad + \lambda^{kd} \left[(1 - \delta - g_L - g)^{\Delta t} + 1_{id > 0} \chi \frac{i d^2}{k d^2} \Delta t \right] + \lambda^S P_{kd} \Delta t \\
v_{ks} = \frac{\partial \mathcal{L}_t}{\partial k s_t} &= \lambda^{ks} (1 - \delta - g_L - g)^{\Delta t} + \lambda^S P_{ks} \Delta t \\
v_S = \frac{\partial \mathcal{L}_t}{\partial S_t} &= c^{-\eta} \left\{ \underbrace{-y \gamma \zeta^2}_{y_S} S \right. \\
&\quad + \underbrace{\frac{\overbrace{\alpha y \varphi_E}^{y_E E}}{y_{base}} \underbrace{E b a s e^{\frac{\epsilon_f - 1}{\epsilon}}}_{E^{\epsilon_f}} E}_{y_\epsilon} \left[-\frac{\ln(E b a s e)}{\epsilon^2} + \frac{1}{\epsilon} \frac{\varphi_c k c^\epsilon \ln(k c) + \varphi_d (k d + \xi)^\epsilon \ln(k d + \xi)}{\underbrace{\varphi_c k c^\epsilon + \varphi_d (k d + \xi)^\epsilon}_{E b a s e}} \right] \\
&\quad \underbrace{\frac{1}{\sigma_0} (1 - \omega) \varrho M^{-\varrho - 1}}_{\epsilon_M} \left(-\frac{1}{M_0} \right) \left. \right\} \Delta t \\
&\quad + \lambda^S.
\end{aligned}$$

Note that the above equations hold in each period. They will be evaluated at time $t + 1$ and for particular realizations of dW^x in the expectations of the loss function.

A few tricks are required to make the algorithm converge and help with reduce the errors.

To have input and outputs of the neural net close to 1, we scale S, λ^S, id, ic, v , by a factor $\zeta, 1/\zeta, 100, 10$ and 0.01 respectively. For S and λ^S this boils down to using temperature as a state rather than S .

We add 4 transformations of the states as extra auxiliary inputs in the neural net: y_{kf}, y_{kc}, y_{kd} and y_S .

To avoid explosive adjustment costs at capital levels close to zero, the adjustment cost are set at a maximum

of 70% of the investment. This constraint will never be binding on an optimal path as this threshold was used to calibrate χ .

We use the first three first order conditions to substitute out the shadow prices of final, clean and dirty capital. For example, for final capital we use

$$c^{-\eta} \left(1 - 2\chi \frac{if}{kf} \right)^{-1} = \lambda^{kf}, \quad (12)$$

and the same for kc . Similarly, we substitute out λ^{kd} using the FOC of id ,

$$\lambda^{kd} = \frac{c^{-\eta} (\varsigma(\varepsilon_1 id) + 1_{id < 0} 2\chi_s \frac{id}{kd}) + \lambda^{ks} \varsigma((-id + \varepsilon_2)\varepsilon_1)}{1 - 1_{id > 0} 2\chi \frac{id}{kd}}. \quad (13)$$

The algorithm is based on Azinovic et al. . We start with a randomized neural net as a policy function. A batch of 200 initial states are randomly selected according to a distribution around the initial states. These initial states produce 200 trajectories of 200 years using the policy function and the difference equations. Next stochastic gradient descent is used to improve the parameters of the policy function. We use a neural net with 3 layers, composed of 200, 200 and 100 nodes. We use gelu activation functions, densely connected with a dropout rate of 1%. Optimizer is AdamW, with a learning rate schedule gradually reducing the learning rate from 7e-4 initially, to 1e-6 after 200k episodes. Stochastic gradient descent is applied on minibatches of 200. The AdamW optimizer increases the learning steps for parameters with larger gradient and gradients that have the same sign and magnitude between minibatches.